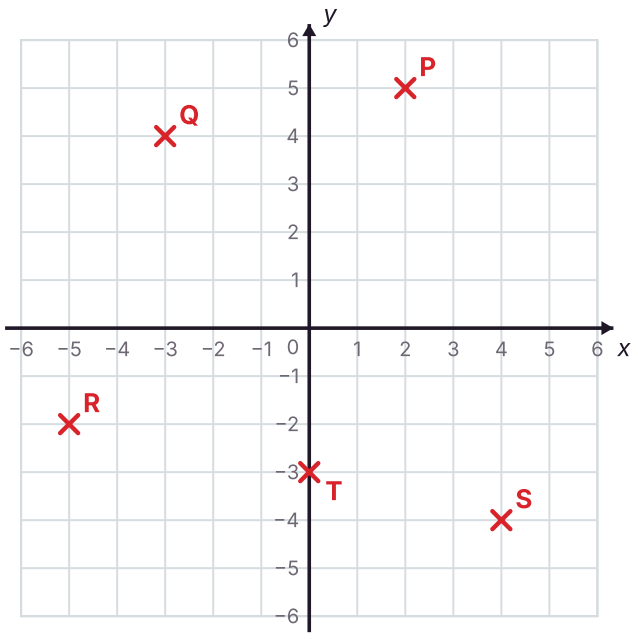
Plotting points – Answer key

Exercise 40.1

1 Read the points

Point	E	F	G	H	I	J
(x, y)	(2, 4)	(−5, 2)	(−4, −3)	(5, −1)	(0, −5)	(−3, 0)
Quadrant	I	II	III	IV	axis	axis



2 Plot and name the quadrant

P (2, 5): quadrant I

Q (−3, 4): quadrant II

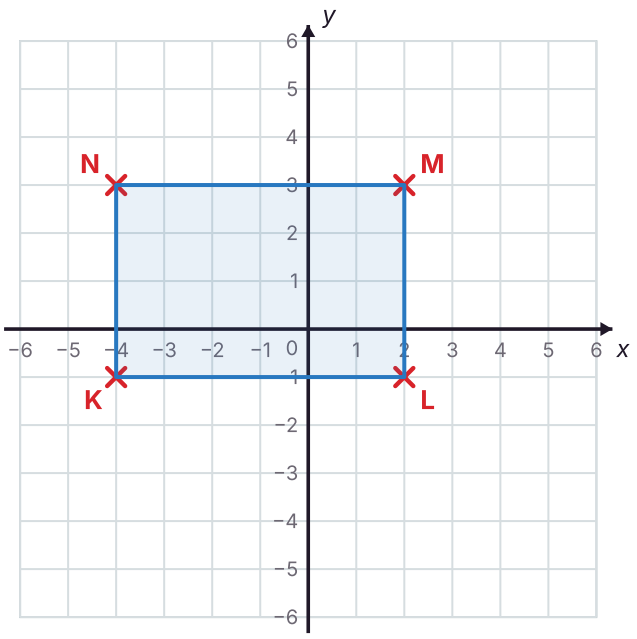
R (−5, −2): quadrant III

S (4, −4): quadrant IV

T (0, −3): on the y-axis, 3 below the origin

3 Ines's mistake

Ines swapped the two numbers. She moved 2 left and 4 up, so she plotted (−2, 4). For (−4, 2) she should move **4 left** first, then **2 up**.



4 The rectangle KLMN

N = (−4, 3).

N is in line with K (same x, −4) and in line with M (same y, 3). The sides of a rectangle are straight up and across.

What do you think?

(3, 2) and (2, 3) are different points: the first number is always along, the second up or down.

If $x = 0$ the point is on the y-axis.

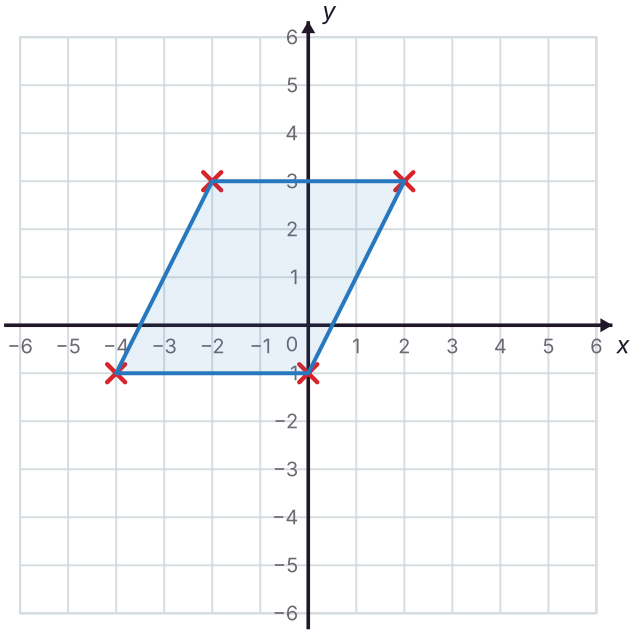
Exercise 40.2

1 Distances from the axes

Point	Distance from the y-axis	Distance from the x-axis
(−5, 3)	5	3
(6, −2)	6	2
(−4, −4)	4	4
(0, 5)	0	5
(−3, 0)	3	0

2 Mystery points

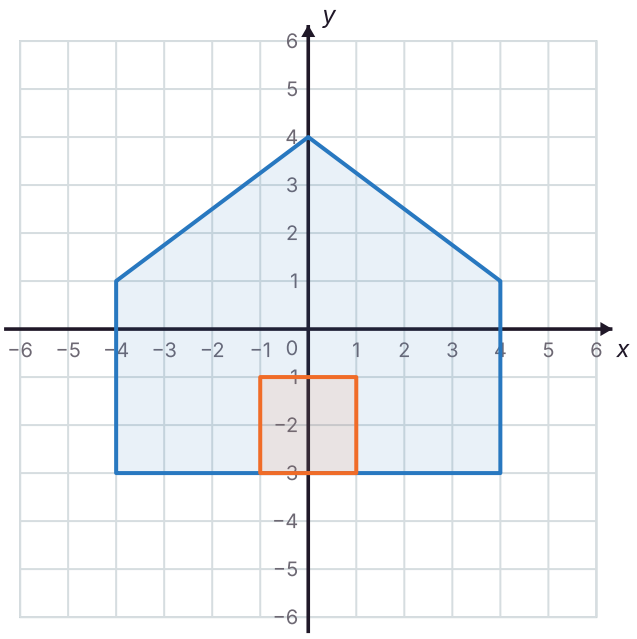
a) (−3, 6) b) (−4, −2) c) (5, 0) and (−5, 0)



3 The parallelogram

The fourth corner is **(−2, 3)**.

From (−4, −1) to (0, −1) is 4 across. Going from (2, 3) 4 across to the left gives (−2, 3).



CHALLENGE

The points joined in order make a **house**: a square with a pointed roof. The second set of points is the **door**.

House: (−4, −3), (−4, 1), (0, 4), (4, 1), (4, −3).

Door: (−1, −3), (−1, −1), (1, −1), (1, −3).